

A Conditional Height-Dominance Architecture for Beal's Conjecture: A Zookeeper-Style Reduction with a Power-Residue Sieve Companion

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Abstract

This paper is a conditional architecture for Beal's conjecture, not a proof. It studies the conjecture along two related lines that share a common arithmetic input.

Part A (deterministic architecture). We formulate a direct Height-Dominance architecture, deliberately modelled on the structural pattern of the Zookeeper-style reduction for the Riemann hypothesis: an intrinsic cluster definition, a cancellation coordinate system, and a three-lemma endgame centred on a coercive complement HD3'. Primitive power structure gives the elementary compression bound $\text{rad}(ABC) < H^{1/x+1/y+1/z}$ ($H = C^z$), and the still-missing Coercive Verifier-Complement HD3' would force a common prime divisor of A, B, C . HD3' is stated as a *Beal-strength local-global conjecture*, not as a weaker technical residue: it concentrates essentially the entire arithmetic content of Beal's conjecture into a single sharply phrased statement, once HD1 (scalar height cancellation) and HD-PC (power compression) have placed any hypothetical counterexample in a controlled height regime. The conditional closure is therefore a *reduction architecture*, not an independent step of progress towards Beal. The opposite-radical-inequality motif appears only as motivation; the actual closure mechanism is the local-global passage HD3'. The open problem is to establish HD3' by arithmetic means.

June 2026 guardrail audits do not upgrade this status: no natural Q97-plus survivor and no direct S-Unit/Frey anchor were found, while synthetic local passes only calibrate negative controls for the root-coherence gate.

Part B (asymptotic sieve companion). For positive primitive solutions with fixed $K \geq 4$ and $3 \leq x, y, z \leq K$, excluding $(3, 3, 3)$, the corrected positive-row reconstruction on classical Jacobi/Weil and additive-large-sieve inputs gives $T_K(H) \ll_{K,\varepsilon} H^{2/3}/\Sigma(Q; K)$ for sufficiently large H and $(\log H)^{2+\varepsilon} \leq Q \leq H^{1/(2K)}$. The lower cut is sufficient, not necessary. The zero integer row is excluded; zero residue rows, empty classes and a zero exclusion budget are handled explicitly. No explicit $H_0(K, \varepsilon)$ or effective constant package is certified here. The formal HD2' entropy budget follows asymptotically from fixed-exponent local densities and primes in fixed arithmetic progressions; it is independent of the bases and proves no root coherence. The exponential bound for the cumulative positive solution count would, under those entropy inputs, force $T_K \equiv 0$; for all fixed K it is already Beal-strength, rather than a weak intermediate estimate.

Version 1.1 corrects the genus and regime classification, the null-row direction and sieve domains, the scalar-family monodromy and strong-sum claims, and the scope of the classical $(3, 4, 5)$ result. The elementary $\Xi \geq \log 4$ inequality provides no IUT/gcd bridge; historical diagnostic scores and hard-coded gates provide no Beal exclusion. Numerical tables remain historical diagnostics. HD3', typed transfers, explicit constants, independent method validation and complete source/proof review remain open.

Keywords: Beal's conjecture, height dominance, radical, power equations, primitive solutions, microcluster architecture, power-residue sieve, large sieve, character sums.

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Part A. Height-Dominance Architecture (Deterministic Route)

1 Purpose

Beal’s conjecture [18] states that every solution of

$$A^x + B^y = C^z, \quad A, B, C, x, y, z \in \mathbb{Z}_{>0}, \quad x, y, z > 2,$$

has a common prime divisor of A, B, C . Equivalently, there is no primitive solution with $\gcd(A, B, C) = 1$.

Part A does not import a proof from the Riemann hypothesis. It borrows only a proof architecture: define the relevant cluster intrinsically, isolate the cancellation, and push the remaining mass or defect into a coercive complement. The overall modular structure of this reduction and verification program is depicted in [fig. 1](#), while the logical core of the three reduction gates is tabulated in [table 2](#).

2 Primitive Power Framework

Definition 2.1 (Primitive Beal counterexample). *A primitive Beal counterexample is a tuple $(A, B, C; x, y, z)$ with $A^x + B^y = C^z$, $x, y, z > 2$, and $\gcd(A, B, C) = 1$.*

Lemma 2.2 (Pairwise separation). *In a primitive Beal counterexample, A, B, C are pairwise coprime.*

Proof. If a prime p divides A and B , then p divides $A^x + B^y = C^z$, hence $p \mid C$, contradicting primitivity. The same argument applies to the pairs (A, C) and (B, C) . $\square$

Definition 2.3 (Height and radical of a solution). *For a primitive tuple set*

$$H_* := C^z, \quad R := \text{rad}(ABC), \quad \sigma := \frac{1}{x} + \frac{1}{y} + \frac{1}{z}.$$

Remark 2.4 (Notation). *Throughout Part A we use H_* for the height of an individual solution, to keep it visually distinct from the global height bound H that appears in Part B’s counting function $T_K(H)$.*

The exponent sum $\sigma = 1/x + 1/y + 1/z$ gives the orbifold Euler characteristic $\chi = \sigma - 1$. Its sign is a signature classification, not an exact genus formula for an unspecified cover. For a genuinely constructed smooth Galois cover of degree d of $\mathbb{P}^1$ with precisely the stated three inertia orders, Riemann–Hurwitz gives $2g - 2 = d(-\chi)$, hence $g = 1 + d|\chi|/2$ in the hyperbolic case. Taking $d = \text{lcm}(x, y, z)$ without constructing the cover is unjustified: $(3, 3, 4)$ would even give $g = 3/2$. The external Darmon–Granville finiteness result for fixed nonzero coefficients and fixed hyperbolic signatures remains separate [5]; it supplies neither emptiness nor a uniform exponent theorem. In the non-cubic Beal domain $\chi \leq -1/12$.

Regime	χ	Examples	Permitted interpretation
Spherical	> 0	$(2, 2, z), (2, 3, 5)$	Signature class; no uniform solution count inferred here.
Parabolic	$= 0$	$(2, 3, 6), (2, 4, 4), (3, 3, 3)$	Cubic Fermat handled separately; models and coefficients matter.
Hyperbolic, mixed	< 0	$(2, 3, 7), (2, 4, 5)$	Fixed-signature external finiteness, without effective emptiness.
Hyperbolic, Beal	$\leq -1/12$	$x, y, z \geq 3$, not $(3, 3, 3)$	Beal remains open; curvature sign is not an exclusion proof.

Table 1: Orbifold signature classification. In particular $(2, 2, z)$ has $\chi = 1/z > 0$ and is not hyperbolic. A cover's genus requires its actual degree and ramification.

3 Power Compression

Proposition 3.1 (Power compression, HD-PC). *Every primitive Beal counterexample satisfies*

$$R \leq ABC < H_*^\sigma.$$

Proof. Since $A^x < C^z = H_*$ and $B^y < C^z = H_*$, we have $A < H_*^{1/x}$, $B < H_*^{1/y}$, $C = H_*^{1/z}$. Thus $ABC < H_*^{1/x+1/y+1/z} = H_*^\sigma$. Since $R = \text{rad}(ABC) \leq ABC$, the claim follows. $\square$

Remark 3.2 (Exponent gap). *The maximum of σ for $x, y, z > 2$ is 1, attained only at $(x, y, z) = (3, 3, 3)$. That case is excluded by Fermat's theorem for exponent 3. For every remaining exponent triple one has $\sigma < 1$.*

4 Power-Radical Microclusters

Definition 4.1 (Power-radical microcluster). *Fix $x, y, z > 2$. The power-radical microcluster $V_{HD}(x, y, z)$ is the data of three disjoint prime packets P_A, P_B, P_C and valuation vectors supported on these packets, producing A^x, B^y, C^z with exponents constrained by the multiples $x\mathbb{Z}$, $y\mathbb{Z}$, and $z\mathbb{Z}$, respectively.*

Remark 4.2 (Non-circularity). *The cluster is defined only from primitivity and the exponent data. It does not assume that the additive equation can or cannot close. That is the endgame statement.*

5 Cancellation Coordinates

Dividing $A^x + B^y = C^z$ by $H_* = C^z$ gives

$$\frac{A^x}{H_*} + \frac{B^y}{H_*} = 1.$$

This is the scalar height cancellation: the leading height term is exactly balanced by the two incoming power terms.

Definition 5.1 (Height defect). *The height defect of a primitive tuple is*

$$D_{HD} := \frac{H_*^\sigma}{R}.$$

By [theorem 3.1](#), any primitive counterexample satisfies $D_{HD} > 1$.

The intended Height-Dominance mechanism would force the opposite inequality, or else force a common prime divisor.

6 The Three-Lemma Endgame

Lemma 6.1 (HD1: scalar height cancellation). *Every primitive solution satisfies $A^x/H_* + B^y/H_* = 1$.*

Proof. This is the original equation divided by $H_* = C^z$. $\square$

Conjecture 6.2 (HD2': Formal Verifier Entropy Budget (Part B form)). *For every fixed $K \geq 4$ there exist constants $A = A(K) \geq 1$, $c = c(K) > 0$, and a threshold $H_0 = H_0(K)$, such that for every primitive power-radical microcluster with exponents $3 \leq x, y, z \leq K$ and $H_* \geq H_0$, the polylogarithmic verifier scale $Q_{\min} = (\log H_*)^A$ satisfies*

$$I(Q_{\min}; x, y, z) \geq c \log H_*,$$

where I is the cumulative verifier entropy of [theorem 11.4](#). In particular, the verifier budget needed to dominate the height H_* grows polylogarithmically in H_* , uniformly in admissible exponent triples.

Remark 6.3 (Corrective status of the formal HD2' budget). *The historical conjecture label is retained for reference, but the displayed existence form follows asymptotically from the fixed- K local-density and prime-progression inputs: $I(Q; x, y, z) \gg_K Q/\log Q$, uniformly over the finite exponent box, and $Q = (\log H_*)^2$ suffices for large height. This budget is independent of A, B, C and does not certify a root section, a beyond-CRT transport, an explicit height threshold or HD3'. The 1000 near-misses are historical diagnostics, not a significance test or an independent validation of the local-global method.*

Conjecture 6.4 (HD3': Coercive Verifier-Complement – Beal-strength local-global form). *If a primitive tuple in $V_{HD}(x, y, z)$ satisfies $V_q(A, B; x, y, z) = 1$ for every prime q , then $\gcd(A, B, C) > 1$.*

Any exact positive primitive closure satisfies all necessary verifier conditions. Thus the displayed HD3' implication, if established with a well-defined tuple/height contract, would imply Beal. No strict stronger/weaker relationship with a finite root-coherence variant is proved here. The pass/fail condition forgets the root and does not bind the otherwise free C ; root identity, height and prime quantifiers require a separate arithmetic contract. This conjecture is not proved.

Remark 6.5 (Status of HD3'). *HD1 and HD-PC are normalisations; HD3' carries the open arithmetic exclusion. Conditional closure is a formal reduction, not a Beal proof. An external theorem about exact $(3, 4, 5)$ solutions does not establish HD3' for every local-pass tuple with an unbound C .*

7 Conditional Closure

Theorem 7.1 (Height-Dominance closure, conditional). *Assume [theorem 6.4](#) for all exponent triples $x, y, z > 2$, with the $(3, 3, 3)$ case handled by Fermat's theorem for exponent 3. Then Beal's conjecture holds.*

The verifier-packet conjecture [theorem 6.2](#) is not a separate logical assumption in this last implication. Its role is diagnostic and constructive: it identifies the local information that any future proof of [theorem 6.4](#) would have to transport beyond the CRT window.

Proof. Suppose a primitive Beal counterexample exists. By [theorems 2.2](#) and [3.1](#), it lies in the power-radical microcluster and satisfies $R < H_*^\sigma$. Since the equality is exact, for every prime q the residue $A^x + B^y \bmod q$ is a z -th power, namely $C^z \bmod q$. Thus $V_q(A, B; x, y, z) = 1$ for every q . [Theorem 6.4](#) then forces a common prime divisor of A, B, C , contradicting primitivity. Hence no primitive counterexample exists. $\square$

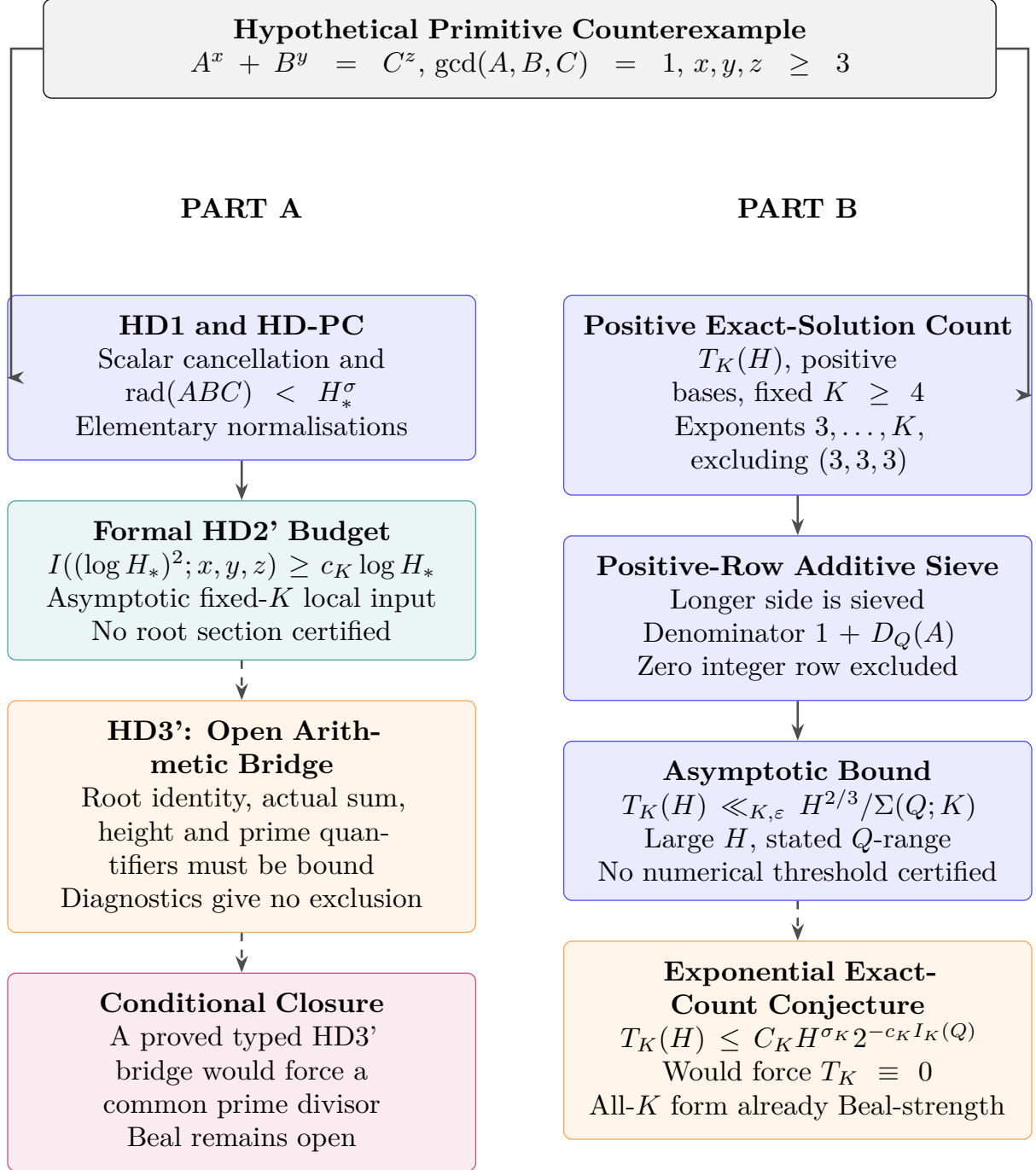


Figure 1: Conditional reduction (Part A) and asymptotic sieve companion (Part B). The formal entropy budget certifies no root coherence. The arithmetic bridge remains open; the polynomial count supplies no deterministic exclusion.

Reduction Gate	Mathematical Claim	Proof / Status Level	Diagnostics & Safeguards
HD1 (Scalar Height Drop)	$A^x/H_* + B^y/H_* = 1$ for primitive solutions	Proven (algebraic identity)	Strict restriction to primitive triples; zero-dominance reference point
HD2' (Formal entropy budget)	$I((\log H_*)^2; x, y, z) \geq c_K \log H_*$ asymptotically	Follows from fixed- K local inputs	Base-independent; no root-coherence certification
HD3' (Arithmetic exclusion)	A typed root/height/prime bridge sufficient for closure	Open hypothesis	Diagnostics and synthetic controls supply no such bridge

Table 2: The algebraic HD1 normalisation, formal asymptotic HD2' entropy budget and open HD3' arithmetic bridge have different roles. Historical diagnostics do not validate the bridge.

8 Approach to HD3': An Honest Sketch

The deterministic route depends on HD3'. Restated: *If a primitive tuple $(A, B, C; x, y, z)$ in the power-radical microcluster satisfies $V_q = 1$ for every prime q , then $\gcd(A, B, C) > 1$.*

What is known

HD1 ([theorem 6.1](#)) is the scalar equation divided by $H_* = C^z$. HD-PC ([theorem 3.1](#)) is the elementary bound $\text{rad}(ABC) < H_*^c$. These are normalisations, not arithmetic exclusion. The existence form of HD2' ([theorem 6.2](#)) follows asymptotically from the fixed- K density and prime-progression inputs; $Q = (\log H_*)^2$ suffices for large height. Its entropy depends on the exponents and primes, not on the actual bases. The historical near-misses are diagnostics and do not establish root coherence.

The hard barrier

A pass says that $A^x + B^y$ has some z -th root modulo q ; it does not identify that root with the chosen C . A valid global exclusion needs a precise contract relating the actual sum, root identity, height and prime quantifiers. Statistical independence of the congruence tests is not justified for the short integer search box. Conversely, this paper proves no general dependence theorem forcing a common prime divisor. The polynomial bound of [theorem 13.1](#) does not give HD3'. The exponential cumulative-count conjecture [theorem 15.2](#) would already imply Beal when formulated for every fixed exponent bound.

Tools that may apply

Classical sieves supply the additive estimate in Part B. Averaging, short character sums, cyclotomic descent, and Frey/S-Unit methods are candidate tools requiring a signature-specific transfer and a uniformity check when fields or prime sets vary. The fixed-signature finiteness theorem of Darmon–Granville [[5](#)] gives no general emptiness certificate. No list of candidate methods constitutes an applicability theorem or a proof that all possible transfers fail.

Concrete sub-goals and their corrected scope

1. **HD3'-RC / HD3'-A.** A finite root-coherence or salt-closing contract must specify positive bases, the actual value $A^x + B^y$, a common bounded integer root and the prime/height quantifiers. A large formal root-capacity defect is not a deterministic obstruction on its own. No strict implication hierarchy or uniform exclusion is certified.

2. **HD3'-B: weak input and rejected uniform strong form.** For nontrivial η , all multiplicative characters extended by zero at zero, put $H_x = \{\chi : \chi^x = 1\}$ and $J(\chi, \psi) = \sum_t \chi(t)\psi(1-t)$. The exact consolidation is

$$S_\eta(x, y; q) = (q-1) \sum_{\substack{\chi \in H_x \setminus \{1\}, \psi \in H_y \setminus \{1\} \\ \chi\psi = \eta^{-1}}} J(\chi, \psi).$$

The one-trivial-character terms cancel the axis contributions; the factor $q-1$ must not be omitted. Classical Jacobi bounds [14, Ch. 8] give the weak $O_K(q^{3/2})$ input. For $q \equiv 1 \pmod{12}$, $(x, y) = (3, 4)$ and η of order 12 there is one resonance pair, so $|S_\eta| = (q-1)\sqrt{q}$. This refutes a blanket η -uniform $q^{1+\varepsilon}$ bound for $\varepsilon < 1/2$ even in a coprime mixed case. It does not refute restricted $(3, 4, 5)$ statements.

The scalar family $a^x + b^y = c$ is constant after the finite étale base change $c = t^L$ on $\mathbb{G}_m$, $L = \text{lcm}(x, y)$: $a = t^{L/x}u$, $b = t^{L/y}v$ reduces it to $u^x + v^y = 1$. In characteristic not dividing xy , its geometric monodromy factors through a finite deck group; this particular family cannot have a positive-dimensional full symplectic monodromy group. No trivialisation at $c = 0$, general Frey-family no-go or Xu-equivalence [23, 8, 16] is inferred.

3. **HD3'-C / HD3'-D.** Counts of all local-pass tuples and counts of exact positive solutions are different. No equivalence of these counting/salt/cofactor goals with [theorem 15.2](#) is proved.
4. **HD3'-E: external exact-solution result.** Siksek–Stoll’s Theorem 1.1 [21] resolves the primitive integer equation $u^3 + v^4 + w^5 = 0$ with only trivial solutions. Putting $w = -C$ excludes positive primitive $A^3 + B^4 = C^5$. It does not prove the proposed verifier statement for every local-pass tuple with an unbound C , nor validate our architecture. The corrected asymptotic sieve at $K = 5$, $Q = H^{1/10}$ gives only $T_{(3,4,5)}(H) \ll H^{29/60} \log H$; $Q = H^{1/5}$ is outside the clean uniform range.

What remains open

HD3' and a typed low-root comparison remain open. The standard additive sieve does not produce the proposed exponential cumulative count bound. The latter is already Beal-strength under the formal entropy inputs, as explained in [theorem 15.2](#). Failed candidate routes or restricted source theorems prove no impossibility theorem for all methods.

9 What Is Not Used

No abc, Szpiro or Riemann-hypothesis implication is used to close Beal. The Zookeeper comparison is architectural. A shared geometric or operator setting for a Beal transfer has not been constructed here; this absence is not a general impossibility theorem for a Hilbert–Polya route or for common geometry.

Part B. Power-Residue Sieve Companion (Asymptotic Bound)

10 The Sieve Setup

Part B supplies a uniform asymptotic counting estimate for the finite exponent box, on named classical inputs. It complements the qualitative fixed-signature finiteness theorem [5], but certifies no explicit height threshold. The entropy table is historical descriptive evidence. The exponential positive-count conjecture is already Beal-strength; it is not an established weaker bridge to HD3'.

11 Power-Residue Verifiers (Part B)

Let q be any prime and $z \geq 3$. Set

$$R_z(q) = \{u^z \bmod q : u \in \mathbb{F}_q\}, \quad |R_z(q)| = 1 + (q-1)/\gcd(z, q-1).$$

Definition 11.1 (Local acceptance density). *For a prime q with $\gcd(z, q-1) > 1$ and exponents $x, y, z > 2$,*

$$\delta_q(x, y, z) := \frac{|\{(a, b) \in \mathbb{F}_q^2 : a^x + b^y \in R_z(q)\}|}{q^2}.$$

For trivial q (i.e. $\gcd(z, q-1) = 1$) the sieve gives no information, so density and entropy sums retain only nontrivial primes.

Definition 11.2 (Verifier indicator). *For a tuple (A, B, x, y, z) and any prime q ,*

$$V_q(A, B; x, y, z) := \mathbf{1}\{(A^x + B^y) \bmod q \in R_z(q)\}.$$

At trivial primes $\gcd(z, q-1) = 1$, $R_z(q) = \mathbb{F}_q$ and $V_q = 1$. Density sums retain only nontrivial primes.

Lemma 11.3 (Necessary condition). *If $(A, B, C; x, y, z)$ satisfies $A^x + B^y = C^z$, then*

$$V_q(A, B; x, y, z) = 1$$

for every prime q .

Proof. $A^x + B^y \equiv C^z \pmod{q}$, and $C^z \bmod q \in R_z(q)$ by definition. $\square$

Definition 11.4 (Cumulative verifier entropy). *For a verifier bound Q and exponents $x, y, z > 2$,*

$$I(Q; x, y, z) := \sum_{q \leq Q, \gcd(z, q-1) > 1} -\log_2 \delta_q(x, y, z).$$

12 Worst-Case Acceptance Density

Fix an integer $K \geq 4$. For every prime with a nonempty eligible index set define

$$\delta_{q,K}^* = \max_{\substack{3 \leq x, y, z \leq K \\ \gcd(z, q-1) > 1}} \delta_q(x, y, z).$$

Primes with an empty index set are omitted from sums using this quantity; an empty maximum is not assigned a numerical value.

Lemma 12.1 (Uniform character-sum bound for δ_q). *For $3 \leq x, y, z \leq K$, $q > K^4$, $q \nmid xyz$, and $d = \gcd(z, q-1) > 1$, the classical Jacobi-sum input gives*

$$\left| \delta_q(x, y, z) - \frac{1}{d} \right| \leq \frac{2K}{\sqrt{q}}.$$

Proof. Write $H_m = \{\chi : \chi^m = 1\}$ for multiplicative characters of $\mathbb{F}_q^*$. Extend every character by zero at zero, including the trivial character. Put $N_0 = \#\{(a, b) : a^x + b^y = 0\}$ and $S_\eta = \sum_{a,b} \eta(a^x + b^y)$. The power-residue indicator yields the exact identity

$$q^2 \delta_q = \frac{q^2}{d} + N_0(1 - 1/d) + \frac{1}{d} \sum_{\substack{\eta \in H_z \\ \eta \neq 1}} S_\eta.$$

For $a = 0$ only $b = 0$ contributes to N_0 ; for $a \neq 0$ there are at most K choices of b . Thus $N_0 \leq 1 + K(q - 1) \leq Kq$.

For nontrivial η , convolution and character orthogonality give

$$S_\eta = (q - 1) \sum_{\substack{\chi \in H_x \setminus \{1\}, \psi \in H_y \setminus \{1\} \\ \chi\psi = \eta^{-1}}} J(\chi, \psi).$$

Indeed, the power-map fibre count is $\sum_{\chi \in H_x} \chi(u) + \mathbf{1}_{\{u=0\}}$. For $u + v = s \neq 0$, substitution $u = st, v = s(1 - t)$ leaves $\sum_{s \neq 0} (\chi\psi\eta)(s)$, which is zero except in resonance. The resonant terms with one trivial character have Jacobi sum -1 ; they cancel exactly against the corresponding axis contributions. The pair with both characters trivial has no resonance because $\eta \neq 1$.

There are at most K remaining resonant pairs, and their product $\chi\psi = \eta^{-1}$ is nontrivial. The classical Jacobi-sum identity gives $|J(\chi, \psi)| = \sqrt{q}$ [14, Ch. 8]. Therefore $|S_\eta| \leq K(q - 1)\sqrt{q}$. Dividing the exact identity by q^2 gives an error at most $K/q + K/\sqrt{q} \leq 2K/\sqrt{q}$. $\square$

Remark 12.2 (Scope of the input). *The argument uses the named classical Jacobi-sum identity, with the character and boundary conventions displayed above. It supplies a local density estimate. It supplies neither a uniform $q^{1+\varepsilon}$ bound for all twisted two-variable sums nor a global exclusion theorem. Historical companion notes and their review seals are retained as historical evidence.*

Lemma 12.3 (Worst-case upper bound). *For eligible primes $q > K^4$,*

$$\delta_{q,K}^* \leq \frac{1}{2} + \frac{2K}{\sqrt{q}}.$$

Proof. For every eligible triple, $d \geq 2$, so $1/d \leq 1/2$. Apply [theorem 12.1](#) and then take the maximum. $\square$

13 The Asymptotic Sieve Bound

Theorem 13.1 (Uniform asymptotic sieve bound). *Fix an integer $K \geq 4$ and $\varepsilon > 0$. Let $T_K(H)$ count primitive tuples with positive integer bases A, B, C , $A^x + B^y = C^z$, $3 \leq x, y, z \leq K$, $C^z \leq H$, and $(x, y, z) \neq (3, 3, 3)$. Define*

$$\Sigma(Q; K) = \sum_{\substack{q \leq Q \\ \exists 3 \leq z' \leq K: \gcd(z', q-1) > 1}} \frac{1 - \delta_{q,K}^*}{\delta_{q,K}^*}.$$

On the classical character-sum, additive large-sieve, and fixed-modulus prime-number-theorem inputs specified below, for sufficiently large H and every

$$(\log H)^{2+\varepsilon} \leq Q \leq H^{1/(2K)}$$

one has

$$T_K(H) \ll_{K,\varepsilon} \frac{H^{2/3}}{\Sigma(Q; K)}.$$

The lower cut is sufficient, not necessary. This statement certifies neither explicit implied constants nor an explicit numerical threshold $H_0(K, \varepsilon)$ or a claimed order $K^{\Theta(K)}$.

Remark 13.2 (Domain and parameter range). *For fixed K, ε the displayed range is nonempty for large H . Small heights admit the trivial positive-box bound; they are not silently included in the asymptotic comparison. For $K = 3$ the sole triple is removed by the classical cubic Fermat case, and $T_3 = 0$; no empty maximum is used. The uniform upper cut absorbs Q^2 into the longer side of the search box.*

Lemma 13.3 (Positive-row additive large sieve). *For M positive rows, let the variable in each row range over an interval of L consecutive integers. For every prime $q \leq Q$ let the allowed row classes have density $\delta_q(A)$, and put*

$$D_Q(A) = \sum_{q \leq Q} \frac{1 - \delta_q(A)}{\delta_q(A)}.$$

If an allowed set is empty, that row has no survivors. Otherwise the total number of survivors is at most

$$\sum_{A=1}^M \min \left\{ L, \frac{L-1+Q^2}{1+D_Q(A)} \right\}.$$

In particular the expression is defined when $D_Q(A) = 0$.

Proof. Fix one row and let Z be its survivor count. Write $S(\alpha) = \sum_n \text{surviving} e^{2\pi i n \alpha}$. Parseval and Cauchy–Schwarz on the allowed residue classes yield

$$\sum_{a=1}^{q-1} |S(a/q)|^2 \geq Z^2 \frac{1 - \delta_q(A)}{\delta_q(A)}.$$

The frequency zero and the reduced fractions a/q for distinct primes are distinct modulo one and Q^{-2} -separated. The classical additive large sieve [3, 24] therefore gives

$$Z^2(1 + D_Q(A)) \leq (L - 1 + Q^2)Z.$$

For $Z > 0$ divide by Z ; for $Z = 0$ the conclusion is immediate. Also $Z \leq L$. Sum over the positive rows. The precise additive input is stated in [24, Theorem 18.1]. $\square$

Remark 13.4 (Row densities and comparison with Σ). *For the classes $a^x + b^y \in R_z(q)$ put $d_y = \gcd(y, q-1)$, $d_z = \gcd(z, q-1)$. The zero-residue row has the exact density*

$$\delta_q(0) = \frac{1 + (q-1) \gcd(d_y, d_z)/d_z}{q}.$$

Thus $\delta_q(0) = 1$ exactly when $d_z \mid d_y$. For example $q = 7, y = 3, z = 6$ gives $4/7$, whereas $y = 6, z = 3$ gives 1 .

For $a \neq 0$ and $q \nmid y$, the polynomial $a^x + b^y$ is squarefree. The classical one-variable multiplicative character bound gives

$$\left| \delta_q(a) - \frac{1}{d_z} \right| \leq 2Kq^{-1/2}.$$

Here the zeros of the polynomial contribute at most K/q , and the nontrivial character sums contribute at most $K/\sqrt{q}$. Consequently, for nontrivial primes $d_z > 1$ and $q > 144K^2$, $\delta_q(a) \leq 2/3$ and the sieve weight is at least $1/2$. At trivial primes $d_z = 1$, the density is 1 and the weight is zero; those primes must not be used for this lower bound.

For a positive integer row A , only its prime divisors have $A \bmod q = 0$. With $\pi_z(Q) = \#\{q \leq Q : \gcd(z, q-1) > 1\}$, this gives

$$D_Q(A) \geq \frac{1}{2} \{ \pi_z(Q) - \omega(A) - \pi(144K^2) \}.$$

The row $A = 0$ is excluded; $\omega(0)$ is not defined. The elementary bound $\omega(A) \leq \log_2 H$ for $1 \leq A \leq H$ is sufficient here, but not asymptotically sharp: if $\omega(A) = k$, then $A \geq (k+1)!$, yielding $\omega(A) = O(\log H / \log \log H)$.

For fixed $3 \leq z \leq K$, primes $q \equiv 1 \pmod{\ell}$ for a prime divisor ℓ of z are nontrivial for z . The fixed-modulus prime number theorem in arithmetic progressions [24, Theorem 4.12] gives

$\pi_z(Q) \asymp_K Q/\log Q$ as $Q \rightarrow \infty$. The row denominator uses this fixed- z prime count, not the larger union of eligible primes for different z' . For $K \geq 4$, $z' = 4$ makes every odd prime eligible, and [theorems 12.1](#) and [12.3](#) imply $\Sigma(Q; K) \asymp_K Q/\log Q$: for large primes each local density lies between a positive K -dependent constant and $2/3$; the finitely many smaller primes have finite weights.

The sufficient lower cut in [theorem 13.1](#) therefore makes $\pi_z(Q)$ dominate $\log_2 H + \pi(144K^2)$ uniformly in the positive rows, for sufficiently large H . No comparison with the upper endpoint $H^{1/(2K)}$ is substituted for the actual Q . The lower cut is not a necessary condition or an equivalence. If Q grows below that cut, $\Sigma(Q; K)$ need not remain bounded; an averaged extension to that range is not established here.

Proof of theorem 13.1. Fix an eligible exponent triple and put $U = H^{1/\min(x,y)}$, $V = H^{1/\max(x,y)}$. The positive (A, B) box has its longer side at most U and its shorter side at most V . Transpose the variables when necessary: the shorter side indexes the positive rows and the longer side is sieved. The row-density argument in [theorem 13.4](#) applies with x, y interchanged as needed.

Throughout the stated range, for large H every positive row has $D_Q(A) \gg_K Q/\log Q \asymp_K \Sigma(Q; K)$. Also $Q^2 \leq H^{1/K} \leq U$. Taking $L = \lfloor U \rfloor$ and $M = \lfloor V \rfloor$ in [theorem 13.3](#) bounds the survivor pairs by

$$\ll_K \frac{UV}{\Sigma(Q; K)} = \frac{H^{1/x+1/y}}{\Sigma(Q; K)}.$$

Each positive pair has at most one positive integer C satisfying $C^z = A^x + B^y$; retaining only primitive pairs and the condition $C^z \leq H$ can only decrease the count. Since $1/x + 1/y \leq 2/3$, summing over finitely many exponent triples gives the stated $\ll_{K,\varepsilon}$ bound. All implied constants are retained; the proof does not replace an asymptotic inequality by an exact coefficient-one inequality. $\square$

Remark 13.5 (Exceptional primes and the unabsorbed remainder). *For fixed K , omitting the finitely many small primes or primes dividing xyz removes $O_K(1)$ total weight; no assertion that each individual weight is at most one is needed. For a fixed triple, under the same sufficient lower cut and large-height condition, the unabsorbed row estimate is*

$$T_{(x,y,z)}(H) \ll_K \frac{H^{1/x+1/y} + H^{1/\max(x,y)} Q^2}{\Sigma(Q; K)}.$$

The uniform upper cut in the theorem absorbs the second term. Explicit constants, a certified numerical height threshold, and an extension to smaller growing sieve ranges remain separate tasks.

14 Empirical Verification

The historical 1000-row near-miss table reports four nested verifier bounds. Its values are reproduced as descriptive diagnostics; the local-density asymptotic is a separate mathematical input.

Remark 14.1 (Descriptive status). *The table reproduces historical reported values without a new run. The formal fixed- K entropy budget follows from classical local inputs and is independent of the bases. Neither the table nor its nested prime bounds supply a significance test, a root section, an exponential survivor bound or $HD3'$.*

Remark 14.2 (Root-coherence guardrail). *The verifier $V_q(A, B; x, y, z)$ is deliberately a coarse pass/fail condition: it asks whether some local z -th root exists modulo q . An exact Beal closure*

Q	$I(Q)$ median (bits)	$I(Q)/\pi(Q)$	slope _{min} vs. $\log H_*$
251	38.24	0.71	+0.704
503	69.78	0.73	+1.116
1009	126.70	0.75	+1.737
2003	232.75	0.77	+4.289

Table 3: Historical reported verifier entropy for 1000 near-misses with $\log H_* \in [16, 39]$. The four nested bounds give descriptive ratios and slopes. They establish neither convergence, statistical significance nor independent method validation.

requires more. The same small integer C must reduce to compatible local roots for all primes under consideration. Equivalently, for

$$R_q(A, B) = \{c \in \mathbb{F}_q : c^z \equiv A^x + B^y \pmod{q}\},$$

one needs a low-height global section $C \leq H^{1/z}$ through the local sets $R_q(A, B)$, not merely the non-emptiness of each set. The empirical entropy table therefore supports only the pass/fail verifier layer. A future HD3' route would need to keep track of root identity, for instance via the root-coherence capacity framework recorded in the companion proof notes. The June 2026 S-Unit/Frey shadow and Q97-plus synthetic-control audits sharpen the same requirement: Q97-plus local survival alone is never evidence. A future survivor may be read as stronger only if it is Q97-plus, branch-bound from the actual value $A^x + B^y$, and accompanied by a low-height global root section or source-bound S-Unit/Frey data.

15 The Stronger Conjecture

Definition 15.1 (Uniform box exponent and entropy). For an integer $K \geq 4$ define

$$\sigma_K = \max_{\substack{3 \leq x, y, z \leq K \\ (x, y, z) \neq (3, 3, 3)}} (1/x + 1/y) = 2/3, \quad I_K(Q) = \min_{\substack{3 \leq x, y, z \leq K \\ (x, y, z) \neq (3, 3, 3)}} I(Q; x, y, z).$$

These index sets are finite and nonempty; no $K = 3$ maximum is used. The exponent is the ceiling of the particular positive-box estimate in [theorem 13.1](#), not a ceiling theorem for all arithmetic methods. The local-density and fixed-progression inputs imply $I_K(Q) \gg_K Q/\log Q$ for large Q .

Conjecture 15.2 (Exponential cumulative-count bound). For each fixed integer $K \geq 4$ there exist constants $a_K \geq 2$, $c_K > 0$ and $C_K > 0$ such that, for sufficiently large H and $Q = (\log H)^{a_K}$,

$$T_K(H) \leq C_K H^{\sigma_K} 2^{-c_K I_K(Q)}.$$

The count is the cumulative positive primitive exact-solution count defined in [theorem 13.1](#), not a count of local-pass pairs.

Remark 15.3 (Why this is already Beal-strength). The logarithm of the right side is at most

$$\sigma_K \log H - c'_K \frac{(\log H)^{a_K}}{\log \log H} + O_K(1) \longrightarrow -\infty.$$

Thus the bound forces $T_K(H) < 1$ for all sufficiently large H . Since T_K is nonnegative, integer-valued and nondecreasing, it follows that $T_K(H) = 0$ at every height. For all fixed K this excludes every positive primitive mixed signature, with the cubic signature excluded classically.

Diagnostic layer	Sample and range	Observed signal	Interpretation
Verifier entropy	1000 near-misses, $Q \in \{251, 503, 1009, 2003\}$	$I(Q)/\pi(Q) = 0.71\text{--}0.77$	Historical descriptive scaling; no root-coherence validation.
Rough root pilot	250 near-misses, $Q \in \{251, 503, 1009\}$	250/250 empty actual root sets; exact-power control 250/250	Local obstruction appears before root drift; the control checks detector sanity.
Small-block survivor scan	250 near-misses, $Q \in \{7, 13, 31, 61\}$	one drift-without-section case at $Q = 31$; all actual rows fail locally by $Q = 61$	The drift zone is finite and transitional in this sample.
Above-threshold scan	5000 near-misses, checkpoints 31–127	five $Q = 61$ passes without low section; 0/5000 local passes at $Q = 97$	No stable above-threshold survivor; follow-up gates must use source-bound families and controls.
S-Unit/Frey shadow audit	five $Q = 61$ survivors, June 2026	0 Q97-plus survivors; 0 direct S-Unit/Frey anchors	Only method-shadow context remains; no source-bound support for HD3’.
Q97-plus synthetic controls	natural survivors plus registered, shifted, and shuffled controls	natural 0/5 Q97-plus; controls reach $Q = 127$ only through synthetic sections or branch maps	Local pass aggregates alone are negative controls; the triple gate is mandatory.

Table 4: Root-coherence and verifier diagnostic ledger. The rows separate evidence for the additive verifier layer from tests probing low-height root sections. All root-coherence rows are internal diagnostics, not proof evidence for HD3’.

Conversely, Beal makes all these counts zero and hence makes the displayed bound immediate. Under the stated entropy inputs the all- K formulation therefore has Beal strength.

A product of densities is exact for a complete CRT residue domain; it is not automatically a survivor probability in a short box. Independence beyond that window is not justified here. No equivalence with a conjecture for local-pass pairs or a strict hierarchy of root-coherence subgoals is asserted.

16 The CRT Product Window

The multiplicative product heuristic is not false in every range. It is exact in the finite Chinese-remainder window, and this exact statement explains where the barrier begins.

Lemma 16.1 (CRT product window). *Let P be a finite set of primes and let $M(P) = \prod_{q \in P} q$. For each $q \in P$, let $S_q \subseteq \mathbb{F}_q^2$ and put $\delta_q = |S_q|/q^2$. Let $S_M \subseteq (\mathbb{Z}/M(P)\mathbb{Z})^2$ be the CRT product of the S_q . Then*

$$|S_M| = M(P)^2 \prod_{q \in P} \delta_q.$$

If $N_P(X, Y)$ counts pairs $(A, B) \in [1, X] \times [1, Y] \cap \mathbb{Z}^2$ whose residue class modulo $M(P)$ lies in

S_M , then

$$N_P(X, Y) = XY \prod_{q \in P} \delta_q + O\left(\prod_{q \in P} \delta_q (M(P)X + M(P)Y + M(P)^2)\right).$$

In particular, relative product asymptotics hold whenever $M(P) = o(\min(X, Y))$.

Proof. The CRT identifies $(\mathbb{Z}/M(P)\mathbb{Z})^2$ with $\prod_{q \in P} \mathbb{F}_q^2$, which gives the exact cardinality of S_M . Each admissible residue class modulo $M(P)$ has $XY/M(P)^2 + O(X/M(P) + Y/M(P) + 1)$ representatives in the box. Multiplying by $|S_M|$ gives the stated formula. $\square$

For the Beal verifier boxes one has $X = H^{1/x}$ and $Y = H^{1/y}$. Thus the direct CRT product window requires

$$M(P) \ll H^{\min(1/x, 1/y)}.$$

For $P = \{q \leq Q\}$, $\log M(P) = \vartheta(Q) \sim Q$, so this direct window reaches only

$$Q \lesssim \min(1/x, 1/y) \log H.$$

In that maximal range the product of local densities gives at most a subpolynomial gain of the shape $\exp(-c\pi(Q)) = H^{-c/\log \log H + o(1)}$. The desired regime $Q = (\log H)^A$ with $A > 1$ already has $M(P) = \exp((\log H)^A)$, far larger than the ordinary search box.

Consequently, the obstruction behind [theorem 15.2](#) is not merely a missing local character-sum improvement. It is the problem of transporting product information beyond the CRT fundamental domain, for example by bilinear dispersion, Frey-family geometry, Selmer or Mordell–Weil structure, or a genuinely new correlation theorem.

17 Comparison with Darmon–Granville

Darmon–Granville [\[5\]](#) establishes qualitative finiteness of primitive solutions for each fixed hyperbolic signature. Finiteness does not establish emptiness. Part B instead supplies the positive-box counting estimate [theorem 13.1](#), uniform over the finite exponent box for fixed K and sufficiently large height, on the stated classical inputs. It does not supply an explicit constant or certified numerical height threshold. No effectivity advantage over the fixed-signature theorem is claimed without that additional work. The signature classification in [table 1](#) does not itself construct a smooth cover or determine its genus.

18 Outlook: Bounded Negative Tests and Open Transfers

The following are candidate-method diagnostics, not general impossibility theorems and not an independent proof of Beal.

18.1 Strong twisted-sum claims and the additive estimate

The mixed $(3, 4)$ resonance example in [section 8](#) invalidates a blanket $q^{1+\varepsilon}$ estimate for every nontrivial character when $\varepsilon < 1/2$. It does not invalidate every character-restricted estimate for a specific signature such as $(3, 4, 5)$. The weaker estimate in [theorem 12.1](#) suffices for the additive bound. Improving local sums alone does not supply the missing beyond-CRT transport in [theorem 15.2](#). Historical attempts with Selberg weights, bilinear dispersion, Heath-Brown identities and Halasz-type bounds [\[1, 10, 9\]](#) record unsuccessful transfers within the tested formulations. They prove no universal second-order parity barrier or exclusion of all standard sieve methods.

18.2 Detection, counting and the cubic case

Prime-detection theorems for polynomial values [6, 7, 11, 12, 19] cannot be imported as exact-solution counting results without a typed reduction. The factorisation $A^3 + B^3 = (A + B)(A^2 - AB + B^2)$ prevents a direct use of a hypothesis requiring an irreducible cubic form; it does not prove categorical incompatibility of detection and counting. The positive primitive (3, 3, 3) case is excluded by classical Fermat for exponent three. Since its count is zero, any upper bound with a nonnegative right side is automatically valid; it is incorrect to say that descent cannot provide such an upper bound. No unbound variable-count threshold is used as a general no-go theorem.

18.3 Signature classification and its limits

The sign of $\chi = 1/x + 1/y + 1/z - 1$ classifies orbifold signatures. In the domain $x, y, z \geq 3$, the only zero case is (3, 3, 3); the other cases are hyperbolic. Outside that domain, (2, 2, z) is spherical, since $\chi = 1/z > 0$. The sign alone does not establish parametric primitive families, an exact smooth-cover genus, sieve applicability or emptiness. Darmon–Granville gives fixed-hyperbolic-signature finiteness. Selmer’s diagonal-cubic examples [20] are a warning about local-global transfer, not a theorem about the Fermat cubic.

18.4 A typed beyond-CRT research question

Can one transfer the product information in [theorem 16.1](#) to a specified short-box count by a proved correlation theorem? A proposed Frey, Selmer, determinant-method or geometric-sieve route [4, 2, 13, 17] must name its objects, map, height comparison, multiplicity bound, family restrictions and uniform constants. No automatic transfer from those cited frameworks to general mixed Beal signatures is established here. In particular, neither a universal $H^{2/3}$ roof nor a sufficient Sato–Tate/Lang–Trotter implication is proved. For the cumulative exact count, the exponential target is already Beal-strength; for a local-pass count it is a different conjecture.

The most recent internal root-coherence notes sharpen this empirical step into a rough-modulus root-coherence test. The diagnostic does not merely count local pass/fail events; it separates small-prime obstructions, rough-modulus density of the root sets $R_q(A, B)$, unit/valuation/branch coherence, and the remaining noncoherent tail. This is an experimental ledger, not a theorem. Its purpose is to make visible whether a low-height global root section fails because of ordinary local obstructions, insufficient rough-modulus density, or a genuinely nonlocal coherence defect.

A limited internal pilot of this ledger on May 22, 2026, using 250 near-miss rows and $Q \in \{251, 503, 1009\}$, found local obstruction before drift in all actual near-miss cases: for each of the three prime blocks, 250/250 actual near-misses had an empty local root set and 0/250 low-height sections survived. June 2026 follow-up audits did not reverse that negative pattern. The S-Unit/Frey shadow audit found 0 Q97-plus survivors and 0 direct source-bound S-Unit/Frey anchors among the five natural $Q = 61$ survivors. The Q97-plus synthetic-control audit showed that local Q97-plus or $Q = 127$ pass aggregates can be manufactured by registered sections or shuffled branch maps, but those controls break branch binding or the low global section. This does not close HD3’. It changes the next empirical task: a positive survivor must satisfy all three gates at once—Q97-plus local survival, actual branch binding from $A^x + B^y$, and a low global root section or source-bound S-Unit/Frey data—before it can be read as more than a diagnostic stress test.

Remark 18.1 (Status of the research question). *These historical tests supply a bounded diagnostic ledger. They neither certify HD3’ nor rule out all alternative methods. The corrected additive theorem retains precisely the domain and large-height conditions of [theorem 13.1](#).*

19 Conclusion and Honest Status

HD1 and HD-PC are elementary normalisations. The formal fixed- K HD2' budget follows asymptotically from the stated classical local inputs and does not certify root identity or global exclusion. HD3' remains open and requires a typed root/height/prime contract. The local density proof, positive-row additive sieve and fixed-progression estimate give [theorem 13.1](#) only in its stated large-height domain. No explicit constant package or certified numerical threshold is supplied. The exponential bound for the cumulative positive primitive exact count would force the count identically to zero; its all- K formulation is already Beal-strength. Historical numerical tables, fitted scores, hard-coded gates and source-file hash matches provide no additional exclusion theorem. This corrective version does not promote the historical claim register, certify an external computational proof, or establish a general impossibility theorem for alternative methods.

20 Corrective Inventory Note for Version 1.1

This section records the scope of the R25 inventory corrections. It separates elementary identities, historical diagnostics, source-reported literature and still-open arithmetic transfers. Original producer files, result tables, claim statuses and historical review seals are not rewritten by this manuscript revision.

20.1 Elementary height identities and diagnostic gates

For an exact positive sum $U + V = H$, put $u = U/H$ and

$$\Xi = \log \frac{H^2}{UV} = -\log[u(1-u)] \geq \log 4, \quad \Delta = \Xi - \log 2 \geq \log 2.$$

These inequalities are elementary and hold for nonprimitive exact tuples as well. Fixed unequal proportions do not make Ξ diverge; divergence needs $\min(U, V)/H \rightarrow 0$. The exact family $(A, B, C; x, y, z) = (3t^5, 6t^5, 3t^3; 3, 3, 5)$, $t \geq 1$, has $u = 1/9$ and constant Ξ , and is nonprimitive. It is no Beal counterexample. An IUT/max-hull or gcd exclusion requires a separate, presently missing arithmetic bridge; the elementary Δ inequality does not provide it. A term defined as $\log H |\chi|$ grows by its definition, not because an arithmetic obstruction was established.

The August/September diagnostic formula $S_{\text{PH}} = -\sum_q (1/q)/(1 + \phi_q^2)$ is negative for every nonempty prime set and every real input ϕ_q . Its sign cannot distinguish a primitive counterexample. The historical branch-binding gate is a constant true flag, not an implemented binding check. The historical survivor predicate uses primitivity, exponent scope and exactness; it does not consume the three diagnostic gates. Its zero survivor count therefore does not validate the proposed exclusion mechanism.

The 25-row historical replay contains eleven primitive in-scope rows: two parabolic (3, 3, 3) rows and nine hyperbolic rows. The named ten-row hyperbolic-primitive group contains one parabolic row. One of five anchored controls has exponent two and is not exact. Accordingly 0/11, 0/10 and 4/5 are different denominators and scopes, not interchangeable success certificates. Nested prime bounds reuse observations; an η -coefficient of variation is descriptive, not a replicated experiment, confidence interval or significance test.

20.2 The July function and provenance boundary

The separate July diagnostic has the form

$$F(w) = \sum_{q,r} \frac{a_{q,r}}{w - e^{2\pi i r/q}}, \quad a_{q,r} > 0,$$

over nonempty root sets. One off-pole evaluation at $w = 0.5 + 0.5i$ does not establish the Herglotz property, which requires holomorphy on the upper half-plane. For the exact nonprimitive control $2^3 + 2^3 = 2^4$ the roots modulo five include $r = 1, 2, 3, 4$, giving upper-half-plane poles with positive residues that do not cancel. This invalidates that Herglotz claim; it does not transfer the separate August sign argument to July.

The inventory found five PDF hashes matching the JSON validation manifest and the checked local files, but conflicting hash fields in the companion Markdown validation note. A local byte-identity check is neither an external provenance audit nor a proof review. Historical files and seals remain preserved; this corrective text does not silently reveal them.

20.3 Bounded literature update

Chocian’s preprint [25] reports a computer-assisted exclusion of nonzero coprime integer solutions of signature $(3, 5, 7)$. Its arXiv identity is confirmed. Its proof and replay companion have not been independently validated here, and it does not certify this paper’s general verifier architecture.

Sahoo [27, Theorem 1.10, Proposition 1.12] is conditional on Conjectures 2.4 and 2.5: for squarefree $d \geq 5$, $-d \not\equiv 1 \pmod{8}$, coefficients $\{\pm 2^r\}$ and the source’s $\alpha_{\mathfrak{p}}\beta_{\mathfrak{p}}\gamma_{\mathfrak{p}} \neq 0$, it excludes sufficiently large *common prime* exponents in $W_{L,\mathfrak{p}}$, where $L = \mathbb{Q}(\sqrt{-d})$, $\mathfrak{p} \mid 2$ and $\mathfrak{p} \mid abc$. The density $5/6$ is relative to squarefree d . This is source-reported scope, not a general mixed-exponent HD3’ theorem or a uniform tower/coercivity estimate.

Pasten [26, Corollaries 1.2–1.3] reports $h(E) \ll N \log \log N$ for elliptic curves over $\mathbb{Q}$ and $h(E) \ll_S N$ for curves semistable outside fixed S . These statements concern the source’s Faltings height and conductor. They neither establish Szpiro’s conjecture nor automatically remove modular forms in a Beal transfer. The required uniform height and S-Unit comparisons when prime sets or fields vary remain open.

A Empirical data and reproducibility

The historical 5000-row near-miss corpus was generated by `evidence/scan_near_power_fractures.py` with `-max-base 250`, `-min-exp 3`, `-max-exp 7` and `-keep 5000`, `-max-rel-gap 1e-4`. The variable- Q experiment retains the first 1000 rows, ordered by relative gap, and uses `evidence/run_variable_Q_sieve.py` for $Q \in \{251, 503, 1009, 2003\}$. Its recorded runtime of about 14 minutes on a two-core server is historical, not a new benchmark.

The curated Beal source and data package is available at <https://github.com/research-line/functional-stability-theory/tree/codex/beal-inventurernte-v1.1/fst-mathematics/beal-height-dominance>. It contains both manuscript sources, the disclosure inputs, eight historical diagnostic scripts, and 17 unchanged CSV files. The package README documents the dataset selection, control classes, replay commands, and claim limits. Its manifest binds the supplied artifacts by SHA-256; the accompanying tests check artifact integrity and the reported finite diagnostic summaries, not a proof of Beal’s conjecture.

The May 22 root-branch-drift pilot supplies three 750-row ledgers: 250 actual near-misses, 250 random-root controls, and 250 exact-power controls for each $Q \in \{251, 503, 1009\}$. The June 1 follow-up uses $Q \in \{7, 13, 31, 61\}$ on the same 250-row selection and the same three control classes. The June 2 rank-8 micro-audit identifies the first empty cubic-root channel at $q = 37$; the 5000-row above-threshold scan has five local passes at $Q = 61$ and none at $Q = 97$. The June 22 S-unit-shadow and June 23 synthetic-control tables are included as historical classifications. Synthetic registered, shifted, or shuffled sections are controls and cannot establish branch binding to the original arithmetic equation. These files support the descriptive audit trail in table 4; they do not establish HD3’, root coherence, statistical significance, or an effective

height threshold. Internal proof and review notes and third-party source PDFs are not part of the public package.

AI Disclosure

AI systems were used extensively in all phases of this work: conception and research discussion, literature and source work, mathematical analysis and proof development, creation and checking of the computational scripts, execution and evaluation of the computational runs, text, translation, and typesetting.

Quality assurance: use of several independent models from different providers with counter-checks and adversarial AI reviews; traceable, versioned computational scripts with stored result files and checksums; continuously maintained proof and register documents; strict claim hygiene (assertions only with documented evidence, addenda instead of silent corrections); changelogs across all versions.

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